

IMSE 586: Big Data Analytics and Visualization

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Project Proposal

**Crime Pattern Analysis and Predictive Policing: A Data-Driven Approach to
Public Safety in Detroit**

Team 01

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Background

Detroit, once the world's auto capital, has long struggled with urban decay and high crime rates stemming from economic decline and population loss. While reports show a downward trend in overall crime in recent years, the city's violent crime levels remain significantly higher than the national average.

Traditional reactive policing is inefficient with scarce funds. Predictive policing offers a data-driven solution, analyzing historical, geo-spatial, and time patterns to forecast crime hot spots. This allows the strategic deployment of limited resources, an approach that has successfully reduced crime in other cities like San Francisco and Chicago.

Research questions

1. What are the temporal crime patterns in Detroit? What months, days, and hours exhibit the highest crime rates?
2. What police districts and geographic clusters are crime hotspots? How does crime density vary across the city?
3. How accurately can statistical and deep learning models predict future crime trends? Which forecasting method provides the most reliable predictions?
4. Can district-level crime forecasts enable targeted resource allocation? Which districts will require increased police in the next 1-5 years?

Literature review

Crime patterns are important to understand to conduct effective prevention measures because crimes are not evenly distributed but rather tend to be focused during certain times and places, referred to as hotspot. Crime hotspots can be broadly interpreted as having greater densities of crime relative to the surrounding areas, with some even necessitating statistically significant clustering[1]. Different theories attempt to explain these patterns, including Routine Activity Theory, which posits that crime is caused by the intersection of motivated offenders, suitable targets, and lack of guardianship[2]. Crime Pattern Theory, which argues offenders commit crimes in places they know according to the effect of crime generators and attractors[3, 4] and Social Disorganization Theory, which blames crime on the characteristics of neighborhoods like poverty, ethnic heterogeneity, and instability. Identification and analysis of such spatio-temporal patterns are essential for police operation briefing and informing urban planning interventions.

Various analytical methods are employed to examine crime patterns. Hotspot mapping, often with techniques like Kernel Density Estimation (KDE), visually brings to the forefront the high densities of crime for closer analysis. Statistical methods such as spatial scan statistics, Local Moran's I, and Getis-Ord G^* test for statistically significant clusters.[1] However, standard density-based clustering methods like DBSCAN can struggle in urban areas with varying crime densities, which suggests that multi-density clustering methods can yield more accurate hotspot detection[5]. Besides, spatial regression analysis models such as Ordinary Least Squares (OLS), Spatial Lag Models (SLM), Spatial Error Models (SEM), and Geographically Weighted Regression (GWR) are used to examine the relationship between the crime rate and the socio-economic or environmental determinants. These spatial models are frequently necessary since crime data typically exhibit spatiotemporal dependence and heterogeneity, which violate the OLS assumptions.

A second key application is the prediction of future crime patterns so that resources can be allocated proactively. Time series prediction using statistical models such as SARIMA, Prophet, HWES, and deep learning models such as LSTM and BLSTM are typically utilized for crime prediction. Studies have applied these techniques at various temporal grains (daily, weekly, monthly) and spatial scales, including city-wide and police district analyses[6]. Comparing the performance of such models, usually based on metrics like Root Mean Squared Error (RMSE) or Mean Absolute Percentage Error (MAPE), is essential to identify the most reliable prediction strategies under specified circumstances. Evaluation of hotspot detection and prediction also incorporates measures like the Hit Rate, Prediction Accuracy Index (PAI), and Density Contrast Ratio (DCR) to assess accuracy and concentration [1], while considering the stability of hotspots over time using measures like the Dynamic Variability Index (DVI) is important for the operational deployment of hotspot policing, especially at micro-temporal time scales[7].

Data set(s)

The [RMS Crime Incidents dataset](#) is a public safety record that includes reports of criminal offenses in the City of Detroit. It contains police department reports from the Detroit Police Department's Records Management System (RMS) that document crimes reported in Detroit where the department was involved. These records cover offenses from December 2016 to present. The data was obtained from the [Detroit Open Data Portal](#), where it was exported from the police department's records system.

Table 1: Detroit Crime Dataset Attributes and Descriptions

No.	Attribute	Description
1	Crime ID	A unique identifier for each incident in the records management system.
2	Report Number	Official report number assigned by Detroit Police Department.
3	Offense Date/Time	Date and time when the offense occurred or was reported.
4	Offense Description	Textual description or code of the crime (e.g., burglary, robbery, assault).
5	Offense Category	Broader classification (violent crime vs. property crime).
6	Location	Street address, block, or approximate location where offense occurred.
7	ZIP Code	Postal ZIP code for the incident location.
8	Police Precinct/District	Police district or precinct responsible for the area.
9	Neighborhood/Area	Named neighborhood or community area in Detroit.
10	X/Y Coordinates	Geographic coordinates (latitude/longitude) for mapping.
11	Year/Month/Day	Derived date parts for temporal analysis.
12	Update Timestamp	When the record was last updated in the dataset.

Method description

- **Data Gathering and pre-processing:** Collect and combine crime incident reports and population data. Clean data by imputing missing values and eliminating duplicates. Geocode exact crime locations. Aggregate crime counts and calculate rates at appropriate temporal (e.g., hourly, daily, monthly) and spatial (e.g., census block groups, police districts) levels.
- **Exploratory Analysis:** Visualize crime distributions over time (year, month, day, hour) and space (hotspot maps, district-level density) to identify basic patterns.
- **Predictive Modeling:** To measure predictive accuracy, we will develop, train, and validate time series forecasting models both statistically (e.g., SARIMA, Prophet) and through deep learning (e.g., LSTM, BLSTM) on aggregate city-wide and police district-level past crime data. Model performance will be contrasted quantitatively through typical metrics (RMSE, MAE, MAPE) upon a withheld test set to decide the best performing forecasting methods.
- **Evaluation:** Assess forecast accuracy using standard metrics (like RMSE, MAE, MAPE) on test data.
- **Application:** Use the best models to generate 1-5 year forecasts and identify high-risk districts to inform resource allocation recommendations.

References

- [1] Timothy C.Hart. “Investigating Crime Pattern Stability at Micro-Temporal Intervals: Implications for Crime Analysis and Hotspot Policing Strategies”. In: *Criminal Justice Review* 46 (2021), pp. 173–189.
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